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1. Real numbers $a_1, a_2, ..., a_n$ are given. For each i $(1 <= i <= n)$ define

   $d_i = (largest\ a_j\ for\ 1 <= j <= i) - (smallest\ a_j\ for\ i <= j <= n)$

   and let

   $d = (largest\ d_i\ for\ 1 <= i <= n)$.

   (a) Prove that for any real numbers $x_1 <= x_2 <= ... <= x_n$, $|x_i - a_i| >= d/2$ for some i with $1 <= i <= n$.

   (b) Show that there exist real numbers $x_1 <= x_2 <= ... <= x_n$ such that equality holds.

2. Consider five points A, B, C, D, E such that $ABCD$ is a parallelogram and $BCED$ is a cyclic quadrilateral. Let l be a line through A. Suppose that l meets segment DC at F and line BC at G. Suppose also that $EF = EG = EC$. Prove that l bisects angle DAB.

3. In a mathematical competition some competitors are friends. Friendship is mutual. A group of competitors is called a clique if each two are friends. Given that the largest clique has even size, prove that the competitors can be arranged in two rooms so that the largest clique size in each room is the same.

4. In triangle ABC the bisector of angle BCA meets the circumcircle again at R, the perpendicular bisector of BC at P, and the perpendicular bisector of AC at Q. Let K and L be the midpoints of BC and AC. Prove that triangles RPK and RQL have equal area.

5. Let a and b be positive integers. Show that if $4ab - 1$ divides $(4a^2 - 1)^2$, then $a = b$.

6. Let n be a positive integer. Let

   $S = \{(x, y, z) : x, y, z \in \{0, 1, ..., n\}, x + y + z > 0\}$.

   Determine the minimum number of planes whose union contains S but does not contain $(0, 0, 0)$.